CSET419 – Introduction to Generative AI

Lab – 10

Objective

The objective of this lab is to implement a generative model capable of learning patterns from sequential data and generating new sequences. Students will explore sequence generation using deep learning architectures such as Recurrent Neural Networks (RNN), LSTM, and Transformer models.

Learning Outcomes

After completing this lab, students will be able to:

1. Understand the concept of sequential data generation
2. Preprocess sequence datasets for deep learning models
3. Implement generative models for sequence prediction
4. Train neural network models to generate new sequences
5. Evaluate the quality and coherence of generated sequences

What is Sequential Data Generation?

Sequential data generation is the process where a model learns patterns from ordered data and generates new sequences that follow similar patterns.

Examples of sequential data include:

- Text sentences
- Speech signals
- Time-series data
- Music compositions
- DNA sequences

Generative models learn the **probability distribution of sequences** and predict the **next element in the sequence** based on previous elements.

Common Approaches

- Markov Models
- N-gram Language Models
- Recurrent Neural Networks (RNN)
- LSTM / GRU Networks

- Transformer Models

Experiment:

Component–I: RNN / LSTM Based Sequential Data Generation

Tasks

1. Load and preprocess the sequence dataset
2. Perform character-level or word-level tokenization
3. Create input-output sequence pairs
4. Design an RNN or LSTM based generative model
5. Train the model on the dataset
6. Generate new sequences using a seed input

Dataset

Use the following sequence dataset:

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artificial intelligence systems learn patterns from data.

sequence models process information step by step.

recurrent neural networks are useful for sequence prediction.

lstm networks handle long term dependencies.


deep learning models improve sequence learning.

generative models create new samples from learned patterns.

language models predict the next word in a sentence.

sequence generation is used in chatbots and assistants.


machine learning helps computers learn automatically.

training data improves model accuracy.

neural networks simulate human brain structures.

optimization algorithms improve learning efficiency.


technology is transforming modern education.
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online learning platforms use artificial intelligence.

students benefit from intelligent tutoring systems.

automation improves productivity and decision making.
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Expected Output

- Generated text sequences.

Component–II: Transformer Based Sequential Data Generation

Tasks

1. Use the same dataset as Component – I
2. Apply word-level or subword tokenization
3. Implement positional encoding for sequence order
4. Design Transformer encoder architecture
5. Train the model for sequence generation

Expected Output

Generated sequences from the Transformer model